

12

Cyber Safety

12.1 INTRODUCTION

Technology breaks barriers and brings minds together in ways that were unthinkable 50 years ago. Our civilization's greatest advancements in recent years have come from technological breakthroughs. Internet is considered as the greatest platform and technology in this century and has become an integral part of our daily lives. It helps us as a learning and communication tool and offers a wide range of opportunities. It is an invaluable source of knowledge and encourages creativity and imagination. Internet ethics imply our behaviour while using it for various purposes.

However, there is a flipside to everything. As information infrastructure and Internet become more complex and much larger, it also becomes critical to maintain the systems up and running all the time with respect to security. Thus, we should be aware of this, be honest, and respect the rights and intellectual property of others on the web.

12.2 CYBER SAFETY

Cyber safety is the safe and responsible use of information and communication technology. It is not only about keeping information safe and secure but also about being responsible with that information, being respectful of other people online, and using 'netiquette' (internet etiquette).

It is a branch of computer technology that is applied to networks and corresponding computers. The objective of this field is to gain knowledge to limit computer crimes, especially those associated with hacking and identity theft, by ensuring security and safety of personal information.

In general, the term cyber safety means the collective mechanism and process by which valuable information and services are protected from publication, tampering, or an assortment of unauthorized activities that are planned and implemented by untrustworthy individuals or unplanned events.



Fig. 12.1: Cyber Safety

CTM: Cyber Safety refers to the safe and responsible use of internet to ensure safety and security of personal information and not posing threat to information available online about others.

To be cyber-safe, there are several strategies which can be adopted to prevent unwanted computer behaviour. Let us discuss them.

12.3 SAFELY BROWSING THE WEB

Protecting yourself by securing your devices, software and connections is important but making the right choices when doing things on the web can make a huge difference to your safety online.

We can ensure our safety while browsing the web by following the tips described below:



(a) Before you start—Update your software

Protect yourself before you start browsing the web by making sure that your operating system, web browser, security software, browser plug-ins (like Java or Adobe products) and other applications are up-to-date.

(b) Protect your web browser (and let your web browser protect you)

You can adjust settings in your web browser to work in a more or less secure way. The functionality might be limited when using the most secure settings but they can provide best protection from malicious content.

Most web browsers will give you warnings when they detect that you are visiting a malicious website or possibly being exposed to malicious content. Pay attention to these warnings. They can help protect you from malware, phishing and identity theft. Learn more about the security settings on your browser.

(c) Observe safe online behaviour

You should observe the following measures when browsing the web to significantly reduce your risk of being a victim of cybercrime:

- Use strong unique passwords online.
- Download files and applications only from those websites that you trust, such as from official app stores or legitimate organizations, such as your bank.
- Pause and think before clicking on links in email, messages or on social networking sites. Don't click on links in messages if you don't know the sender or if the message is unexpected.
- If a link looks suspicious or you can't tell where it leads to, before clicking, hover over that link to see the actual web address it will take you to (usually shown at the bottom of the browser window). If you do not recognize or trust the address, try searching for relevant key terms in a web browser. This way you can find the article, video, or webpage without directly clicking on the suspicious link.
- Beware of offers that seem too good to be true. Leave websites that ask for your personal or banking details in return for money—these are scams.

(d) **Online transaction handling**

While making purchases online:

- Check if the site is reputed and has a refund policy.
- While making payments, check that you are using a secure connection. The URL of a secure payment page will use 'https' instead of 'http' and a padlock icon will be displayed by your browser.
- If the website looks suspicious or you have doubts, do not proceed.
- Malware can be delivered through malicious advertising (known as malvertising). Using an ad blocker can stop malware from being delivered through your browser. Some browsers include an Ad Block feature. Avoid using public computers or Wi-Fi hotspots to access or provide personal information. Don't use online banking or make payments with credit cards using public computers or Wi-Fi.

(e) **Cookies and security**

Cookies are small text files—bits of information—left on your computer by websites you have visited which let them 'remember' things about you.

Websites use cookies in order to gather information about their visitors. Cookies may also be used to store your preferences and settings for particular websites which means your experience can be customized based on your past behaviour.

From security perspective, cookies are unlikely to be used maliciously against you as they are just text read by your browser—they don't contain any code that could be executed. However, websites are able to gather a lot of information about you and what you do on a website using cookies.

If you have concerns about how this might impact your privacy, you could consider regularly clearing the cookies from your computer or device.

(f) **Bookmark important sites**

If there are sites you visit regularly, it's a good idea to bookmark them in your browser. A mistyped address could take you to a false site that mirrors the site you intend to visit but with a malicious code that can harm your computer and compromise your information. Bookmarked addresses take you to the same site every time.

Most importantly, remember that the care you take to protect yourself in everyday life extends to your online life. Don't share your personal information unless you are absolutely certain that it will not be compromised. Be aware that any information you share over a public wireless hot spot (say, at your favourite coffee shop) is not secure and can be seen by anyone looking at information as it travels over that network. Just as you wouldn't leave your debit card on the dashboard of your unlocked car, don't make your information easy to attain.

12.4 IDENTITY PROTECTION WHILE USING INTERNET

Identity theft is a type of fraud that involves using someone else's identity to steal money or gain other benefits. Online identity theft refers to an act of stealing someone's personal information such as name, login details, credit card details, etc., and then posing as that person online. The intent is to use that identity for personal gains, generally with the intent to cheat others.

To protect ourselves from such frauds, we should follow **Private Browsing** or **Anonymous Browsing** on the internet. Private browsing is a type of browsing wherein the browser opens in incognito mode or through proxy or VPN (Virtual Private Network) and does not store cookies.

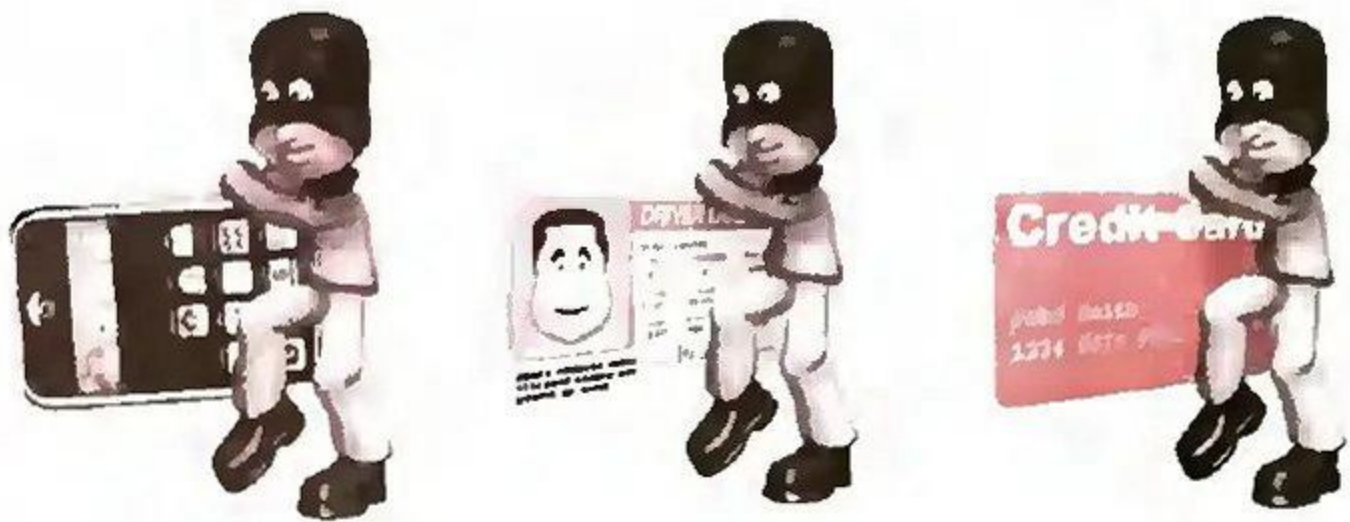


Fig. 12.2: Identity Theft

CTM: Identity theft occurs when someone, without your knowledge, acquires a piece of your personal information and uses it to commit fraud.

12.4.1 Ways of Tracking Your Identity

Whenever you access or visit a website, your web browser may reveal your location through your device's IP address. It can also provide your search and browsing history which can be used by people like advertisers or criminals who intend to use it for some personal gains. In other words, websites track you through various methods as described below.

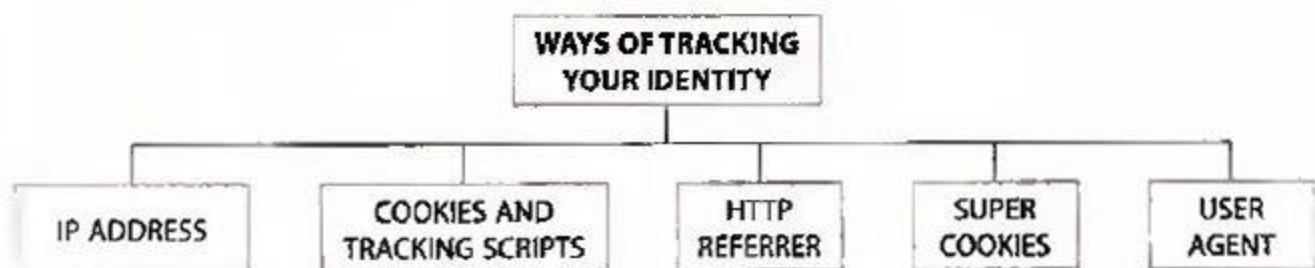


Fig. 12.3: Methods of Tracking Identity

1. **IP Address:** It is a unique address of your device when you connect to the internet. When you share your IP address on the network with other devices, a website can determine your geographical location.
2. **Cookies and Tracking Scripts:** Cookies, as explained earlier, are small text files on your computer storing small piece of information related to your online habits/text. They can identify you and track your browsing activity. Cookies are of two types:
 - (a) **First Party Cookies:** These are cookies that store your own login id, password and auto fill information, etc., for some websites that you frequently visit.
 - (b) **Third Party Cookies:** Cookies that websites store to know about your search history and web browsing history so as to place advertisements as per your interests are known as third party cookies.

3. **HTTP Referrer:** It is an HTTP header field that identifies the address of the web page (i.e., the URL) that links to the resource being requested by the user. By checking the referrer, the new web page can see where the request originated and pass your details such as IP address, your mail id, password, etc., to these linked sites.

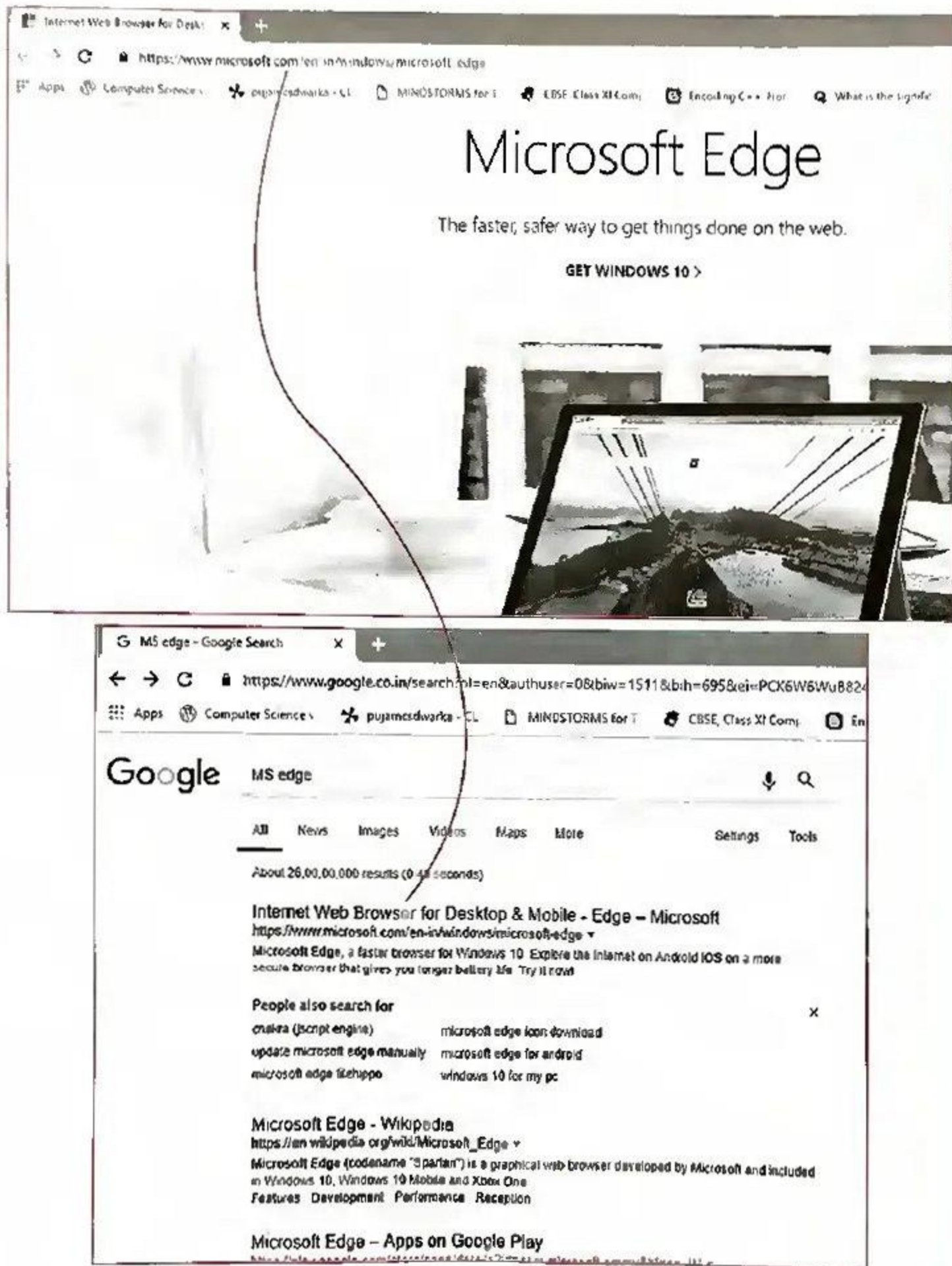


Fig. 12.4: HTTP Referrer

When we click on the link, the browser will send the HTTP REFERRER to the new site. In other words, microsoft.com will know the URL where we are coming from.

4. **Super Cookies:** These are persistent cookies, i.e., they come back even after being deleted. They store cookie data in multiple places, for example, in Flash cookies, Silverlight storage, your browsing history and HTML5 local storage, etc.

When you delete a part of super cookie, this can get easily noticed by a website which can then repopulate the information deleted from some other alternate location. For example, if a user clears browser cookies and not his Flash cookies, in such a case the website may copy the value of the Flash cookies to your browser cookies.

5. **User Agent:** Your browser also sends a user agent every time you connect to a website. This tells websites about your browser and operating system, providing another piece of data that can be stored and used to target ads.

12.5 CONFIDENTIALITY OF INFORMATION

Confidentiality is the term used to prevent disclosure of information to unauthorized individuals or systems.

Data should be kept secret. The owner of the data has to decide who can access the data and who can't. One of the most common threats to confidentiality nowadays is Password Hacking in online money transaction systems.



Fig. 12.5: Confidentiality of Information

CTM: Confidentiality of information ensures that only authorized users get access to sensitive and protected data.

Confidentiality of our important information can be maintained by observing the following practices:

1. Prevention by encrypting the data and by limiting the places where it might appear.
2. Using firewall wherever possible.
3. Controlling browser settings to block tracking.
4. Browsing privately wherever possible.
5. Taking care while posting on Social Media.
6. Ensuring safe sites while entering crucial information.
7. Ensuring that the address contains HTTPS and a padlock sign.
8. Carefully handling emails.
9. Not giving sensitive information on wireless networks.
10. Avoiding use of public computers.

12.6 CYBERCRIME

Any criminal act that is facilitated by the use of electronic gadgets such as computer, smartphone, laptop, etc., involving communications or information systems through internet is referred to as Cybercrime.

12.6.1 Cyber Trolls

A cyber troll is a person who starts quarrels or upsets people on the internet by posting inflammatory, digressive, extraneous or off-topic messages to an online community (such as a newsgroup, forum, chat room or blog) with the intent of provoking readers into displaying emotional responses and normalizing tangential discussion, either for the troll's amusement or for a specific gain.

More often these cyber trolls end up as cyber bullies. So, we can say that cyber trolling is a form of cyberbullying.



Cyber Trolls



Cyberbullying



Cyber Stalking

Fig. 12.6: Common Cybercrimes

12.6.2 Cyberbullying

Cyberbullying is an attack on an individual or a group through the use of electronic means such as instant messaging, social media, email and other forms of online communication with the intent to abuse, intimidate or overpower.

Types of Cyberbullying

- Posting any kind of humiliating content about the victim
- Hacking the victim's account
- Sending or posting vulgar messages online
- Threatening to commit acts of violence
- Stalking by means of calls, messages, etc
- Threats of child pornography

12.6.3 Cyber Stalking

Cyber stalking is the use of internet or other electronic means to stalk or harass an individual, a group of individuals, or an organization.

It may include making false accusations or statements of fact (as in defamation), monitoring, making threats, identity theft, damage to data or equipment, solicitation of minors for objectionable activities, or gathering information that may be used to harass someone.

12.6.4 Spreading Rumours Online

With the widespread use of social networking sites nowadays, people think that they can post anything and everything on these sites by creating fake email ids and masking their actual identity and believe that they will never be caught. Through such fake profiles, they post false information and spread rumours or make comments which may hurt others and result in quarrels, and sometimes even lead to communal tension.

Thus, spreading rumours should be checked and taken care of as it comes under the category of cybercrime and is a punishable offence under the Information Technology Act.

12.6.5 Reporting Cybercrime

In case any cybercrime involving a child or student is detected, it should first be brought to the notice of the parents and school authorities and then reported to the police. The procedure to be adopted is same as in any other kind of crime or offence.

- The local police stations can be approached for filing complaints, which have designated cybercrime cells to register complaints.
- A complaint can also be registered by filing an “e-FIR” which is very convenient. This facility is available in many states.
- Recently, the Ministry of Home Affairs has also launched a website for registering crimes against women and children online including cybercrimes.

An important feature of the Information Technology Act is that a cybercrime has global jurisdiction, which means that the crime can be reported in the cyber cells or cyber units of any city, irrespective of the place the act has been committed.

12.7 CYBER FORENSICS

Cyber forensics is an electronic discovery technique used to determine and reveal technical criminal evidence. It often involves electronic data storage extraction for legal purposes. Although still in its infancy, cyber forensics is gaining traction as a viable way of interpreting evidence.

Cyber forensics is also known as computer forensics. The various capabilities of cyber forensics are:

- Recover deleted files
- Find out what external devices have been attached and who have access to them
- Determine what programs have been run
- Recover web pages
- Recover emails and users who read them
- Recover chat logs
- Determine file servers used
- Discover a document’s hidden history
- Recover phone records and text messages from mobile devices
- Find malware and data collected

12.8 APPROPRIATE USAGE OF SOCIAL MEDIA

The term social media refers to web and mobile technologies and practices that people use to share content, opinion, experiences and perspective online.

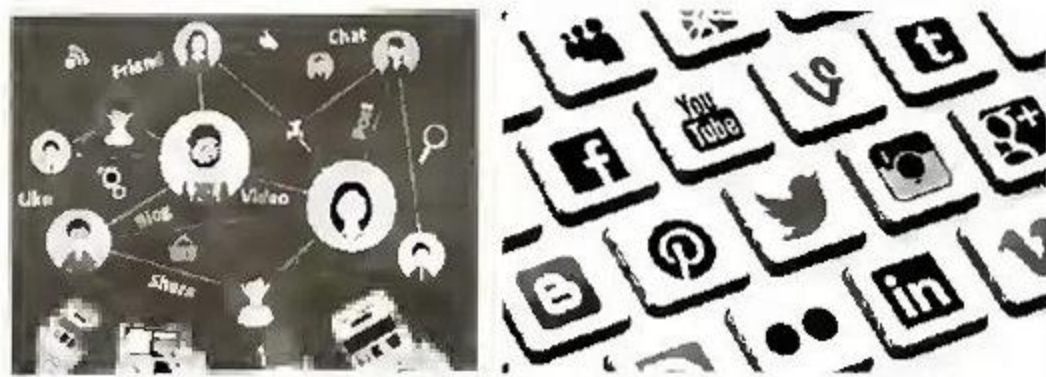


Fig. 12.7: Social Networking Sites

Social networking sites can support participation, collaboration and community building and prove a useful resource for team-based projects, particularly those involving joint problem-solving. The use of social networking sites can extend to peer mentoring, informal student interaction around specific projects, online discussion and collaboration, promotion of events, and pooling of resources.

Some of the prominent social media platforms are:

- ✓  **Facebook**—Facebook is a social networking website where users can post comments, share photographs and post links to news or other interesting content on the web, chat live, and watch short-form videos. Social network services are primarily used for socializing with existing friends.
- ✓  **Twitter**—Twitter is an online news and social networking site where people communicate in short messages called tweets. Tweeting is sending short messages to anyone who follows you on Twitter, with the hope that your messages are useful and interesting to someone in your audience.
- ✓  **YouTube**—YouTube is a video-sharing service where users can share, watch, like and comment on videos. The video service can be accessed on PCs, laptops, tablets and mobile phones.
- ✓  **LinkedIn**—LinkedIn is a social network for the business community. Founded in 2003, the online site is a place for professionals to connect with past and current colleagues, increase their number of business connections, network within their industry, discuss business ideas, search for jobs and look for new hires.
- ✓  **Blogging Sites**—A blog (shortening of “weblog”) is an online journal or information website displaying information in reverse chronological order, with latest posts appearing first. It is a platform where a writer or even a group of writers share their views on an individual subject.

Rules to be followed while working on social networking sites

- Don't give or post any personal information.
- Never give out your password to anyone except your parent or guardian.
- When you are choosing a social networking site, privacy issues should be considered.
- Add only those people as friends to your page whom you know in real life.
- Delete any unwanted messages, inappropriate comments and immediately report those comments to the networking site.
- *Share but with care:* Social media wants us to share as much as we can, but “Never share your passwords, private/personal information, your location, etc.” Always use strong passwords that are difficult to guess.
- *Privacy matters:* Regularly check your privacy settings on social media and always think before posting because it spreads all over the internet.
- *Respect privacy of others:* Ask your friends' permission before uploading their photos and videos.
- *Keep everything updated:* Most security breaches on the internet happen as the software is not up-to-date. Be vigilant about updating software like apps, anti-viruses and browsers.

- *Be aware of Spams:* It is the modern version of junk mails. Learn the difference between real mails and messages and dodgy things with unreliable links. Always be suspicious of emails with hyperlinks.
- *Cyber bullying and harassment:* If you are targeted by this kind of behaviour, you must know what to do and where to get help. Visit National Centre Against Bullying website.
- *Keep your friends close and strangers at arm's length:* Never accept friend requests from strangers.
- *Check your reaction to trolls:* Don't feed the trolls by reacting angrily. In fact, this is what they want. Block/Report the trolls.

12.9 NETWORK SECURITY THREATS

A threat refers to anything that has the potential to cause serious harm to a computer system. Threats are potentials for vulnerabilities to turn into attacks on computer systems, networks and more. They can put individuals, computer systems and business computers at risk. Therefore, vulnerabilities have to be fixed so that attackers cannot infiltrate the system and cause damage.

Network security measures are needed to protect data during their transmission and to guarantee that data transmissions are authentic.

12.9.1 Denial of Service (DoS) Attacks

DoS attacks are those attacks that prevent legitimate users from accessing or using resources and information. These types of attacks may eat up all the resources of the system and bring the computer to a 'halt'. The various DoS attacks are:

- **Denial of Access to Information:** Corrupting, encrypting or changing the status of information so that it is not accessible to its legitimate user.
- **Denial of Access to Application:** Forced shutting of an application as soon as the user opens it.
- **Denial of Access to Resources:** Blocking a resource, may be a printer or scanner or USB port, of a computer, preventing the computer from working properly.
- **Denial of Access to a Website:** Continuously sending bulk requests to a website so that it is not available to any other user.

12.9.2 Malware

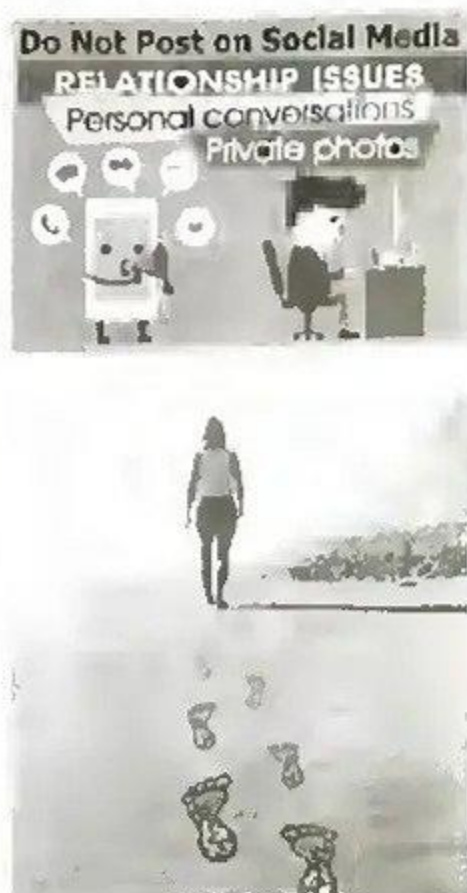
Malware (short for "malicious software") is a file or code, typically delivered over a network, that infects, explores, steals or conducts virtually any behaviour an attacker wants. Though varied in type and capabilities, malware usually has one of the following objectives:

- Provide remote control to an attacker to use an infected machine.
- Send spam from the infected machine to unsuspecting targets.
- Investigate the infected user's local network.
- Steal sensitive data.

Malware is an inclusive term for all types of malicious software, such as:

- **Cyber Bullying:** Cyber bullying is the use of technology to tease, humiliate, threaten and harass someone.
- **Stranger Danger:** Stranger danger is the danger to children and adults presented by strangers.
- **Digital Footprint:** This is used to describe the trail, traces or "footprints" that people leave behind. Digital footprints are recorded with each and every online activity we perform—whether it is interactions on social media, chatting and connecting with friends on social media sites, sites that we visit, online shopping, locations through Facebook check-ins, etc.

These can be compared more often to the footprints that you leave behind when walking along a beach. Every step that you take leaves an impression on the sandy surface that allows another person to see and follow your marks. Digital footprints are also termed as "Digital Tattoos". However, a beach footprint gets washed away after some time whereas digital footprints stay forever and can't be done away with.



12.9.3 Viruses

Viruses are programs that copy themselves throughout a computer or network. Viruses can only be activated when a user opens the program. At their worst, viruses can corrupt or delete data, use the user's email to spread, or erase everything on the hard disk. If not checked, a virus can damage the system or specific files which are important for the functioning of an operating system. It may also slow down your computer as a result of replication of virus in the background without being visible on the screen.

12.9.4 Worms

Worms are self-replicating viruses that exploit security vulnerabilities to automatically spread themselves across computers and networks. Unlike many viruses, worms do not attach to existing programs or alter files. They typically go unnoticed until replication reaches a scale that consumes significant system resources or network bandwidth.

12.9.5 Trojans

Malware disguised in what appears to be legitimate software are known as Trojans. Once activated, Trojans will conduct whatever action they have been programmed to carry out. Unlike viruses and worms, Trojans do not replicate or reproduce through infection.

2.9.6 Zombie Computers and Botnets

In Computer Science, a zombie is a computer connected to the internet that has been compromised by a hacker, computer virus or Trojan horse and can be used to perform malicious tasks under remote direction.

Zeus, Koobface, TidServ, Monkif, Hamwep, Swizzor, Ganminima, etc., are a few examples of zombie computers that have been used for Distributed Denial-of-Service (DDoS) attacks, spam, phishing, information theft and distributing other malware.

Botnets (Short for "robot network") are often used to spread email spam and launch Denial-of-Service attacks. Most owners of zombie computers are unaware that their system is being used in this way.

12.9.7 Spyware

Malware that collects information about the usage of the infected computer and communicates it back to the attacker is known as Spyware. Spyware mostly gets installed on your PC without your consent. It finds its way into PCs by "piggybacking" onto a file, or gets downloaded from the internet when you visit a particular website.

Spyware is used for various purposes. It monitors your internet activity, tracking your login and password information, and spying on your sensitive information. Spyware is not just one type of program. It is an entire category of malicious software that includes adware, keyboard loggers, Trojans and mobile information stealing programs.

12.9.8 Phishing and Pharming

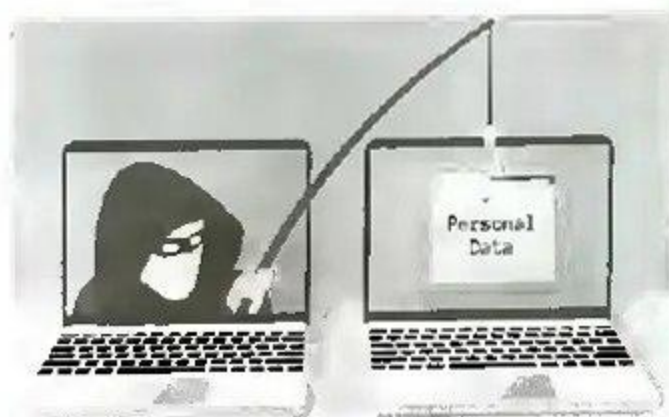


Fig. 12.8(a): Phishing

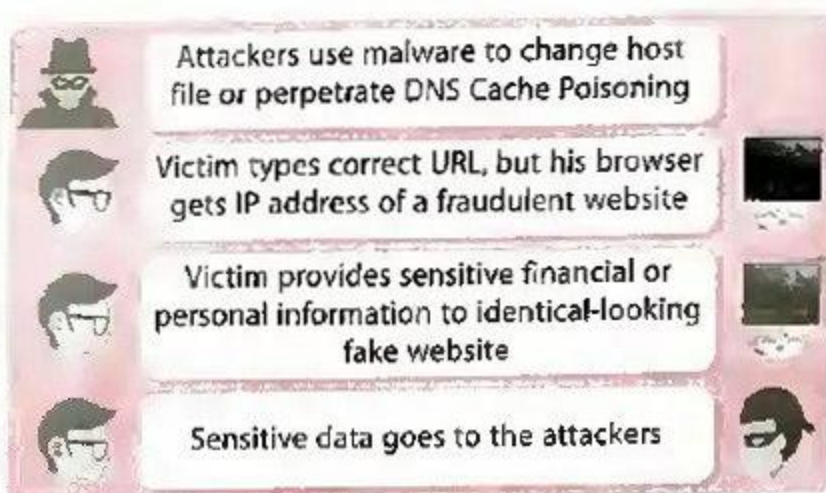


Fig. 12.8(b): Pharming

Phishing is the attempt to acquire sensitive information such as usernames, passwords and credit card details (and sometimes, indirectly, money) by masquerading as a trustworthy entity in an electronic communication.

Phishing often occurs through emails and instant messaging and may contain links to websites that direct the user to share private information.

These fake websites are often designed to look identical to their legitimate counterparts to avoid suspicion.

Pharming is an attack in which a hacker attempts to redirect a website's traffic to another fake or bogus website. Through Pharming attack, the attacker points you to a malicious and illegitimate website by redirecting the legitimate URL. Even if the URL is entered correctly, it can still be redirected to a fake website.

12.9.9 Adware

Adware refers to a type of malware that displays unwanted advertisements on your computer or device. Adware is commonly activated unknowingly when users are trying to install legitimate applications that adware is bundled with.



Fig. 12.9: Adware

Adware commonly takes the form of irritating pop-up windows and banners; however, it can act in many different ways, some of which are as under:

- Acting similarly to spyware, some types of adware track your movement and activities online in order to tailor specific adverts to you.
- Operating as a middleman, adware can redirect your activities in order to share adverts with you.
- Adware uses up your data, with every pop-up download eating away at your allowance.
- Running adware uses up power, affecting your device's performance.

12.9.10 Child Pornography

Child pornography is publishing and transmitting obscene material about children in electronic form. In recent years, child pornography has increased due to easy access to the internet and easy availability of videos on the internet. Child pornography is the most heinous crime and has led to various other crimes such as sex tourism, child abuse, etc.

Child pornography laws provide severe penalties for producers and distributors of such material in almost all Western societies, including incarceration, with shorter duration of sentences for non-commercial distribution depending on the extent and content of the material distributed. However, prison sentences are often converted to probation for first-time offenders.

Safeguards for students:

1. Never share your address with people whom you do not know.
2. Never publish your personal information on public sites/social networking sites.
3. The best safety option on Facebook is the "only friends" option.
4. Do not open suspicious emails.

Tips for parents:

1. Talk with your children about the importance of keeping personal information private.
2. Inform them about identity theft and how they can prevent it from occurring.
3. Set some internet rules pertaining to websites a child is allowed to access.
4. For a price, obtain an identity theft program that protects your computer's privacy.

Tips for educators:

1. Teach students the importance of keeping their personal information private.
2. Monitor the websites that students have access to.
3. Teach students the dangers of viruses and opening suspicious emails.

Complete Network Security Solution:

1. Never open or download a file from an unsolicited email/website.
2. Keep your operating system updated.
3. Use an authentic antivirus program.
4. Enable two-factor authentications whenever available.
5. Confirm the authenticity of a website prior to entering login credentials.
6. Look for HTTPS in the address bar when you enter any sensitive personal information on a website to make sure your data will be encrypted.

12.10 COMMUNICATING DATA SAFELY—SECURE CONNECTION

A secure connection is a connection that is encrypted by one or more security protocols to ensure the security of data flowing between two or more nodes. When a connection is not encrypted, it can be easily listened to by anyone and is even prone to threats by malicious software and rogue and unexpected events.

Anyone who wants to get information from a non-secured connection can do so since they can easily go through the computer's network taking with them important data such as login, passwords and other private information.

Secure connections, as they are supposed to protect data being transferred from one computer to another, must be able to do three main tasks:

1. Prevent third parties from getting hold of confidential data,
2. Validate the identification of the person first who wishes to access and exchange data,
3. Protect information from being viewed or altered by unknown parties.

There are many methods to establish a secure connection but most of them involve data encryption.

Data encryption is a method which hides information from unauthorized parties. This method usually needs an appropriate program installed on both computers involved in the connection that will encrypt and decrypt information. Among these are our basic security protocols embedded in the main communication protocols like TCP/IP, HTTPS, POP3 or IMAP.

Firewalls and antivirus software may also serve in creating secure connections in some form.

➤ Who needs secure connection

The various stakeholders for a secure connection are those who are involved with some sort of online transaction processing such as:

1. *Business industry*

The three main reasons for securing business communication are:

- To protect customer data—health-care companies, businesses in the legal and financial sectors, tax advisors and private banking;
- To protect intellectual property amidst fears of growing industrial espionage, in particular of companies from the pharmaceutical, automotive and industrial sectors;
- To protect internal communication—government institutions and large corporate houses.

2. *Financial and banking industry*

The financial and banking industry is among the primary targets of any crime, including cybercrime. It is literally where the money is. The client information that banks store is also of interest because it could be used to commit identity theft and fraud. To counter these threats, the banking industry needs to secure its communications.

One of the most extensively used mechanisms is to access websites only with "HTTPS" as a prefix for online transaction processing.

HTTP vs HTTPS

When you request an HTTPS connection to a web page, the website will initially send its SSL certificate to your browser. This certificate contains the public key needed to begin the secure session. Based on this initial exchange, your browser and the website then initiate the 'SSL (Secure Socket Layer) handshake'. The SSL handshake involves the generation of shared secrets to establish a uniquely secure connection between yourself and the website.

When a trusted SSL Digital Certificate is used during a HTTPS connection, users will see a padlock icon in the browser address bar. When an Extended Validation Certificate is installed on a website, the address bar will turn green.

All communications sent over regular HTTP connections are in 'plain text' and can be read by any hacker that manages to break into the connection between your browser and the website. This presents a clear danger if the 'communication' is on an order form and includes your credit card details or social security number. With an HTTPS connection, all communications are securely encrypted. This means that even if somebody manages to break into the connection, they would not be able to decrypt any of the data which passes between you and the website.

Steps to check if a site's connection is secure

To see whether a website is safe to visit, you can check for security info about the site. Your browser will alert you if you can't visit the site safely or privately.

For example, if you are browsing using Chrome:

1. In Chrome, open a page.
2. To check a site's security to the left of the web address, look at the security status:
 - Secure
 - Info or Not secure
 - Not secure or Dangerous

Some more facts about a secured communication

In recent years, WhatsApp has added end-to-end encryption, which means only the sender and the recipient of the message can read it due to special software keys that unlock the encryption.

End-to-end encryption is when your service provider doesn't keep copies of the messages you send on their servers. Messaging apps like WhatsApp are Signal, Viber, Telegram, Dust, which are available on platforms like iOS, Android, Windows, macOS and Linux free of cost.

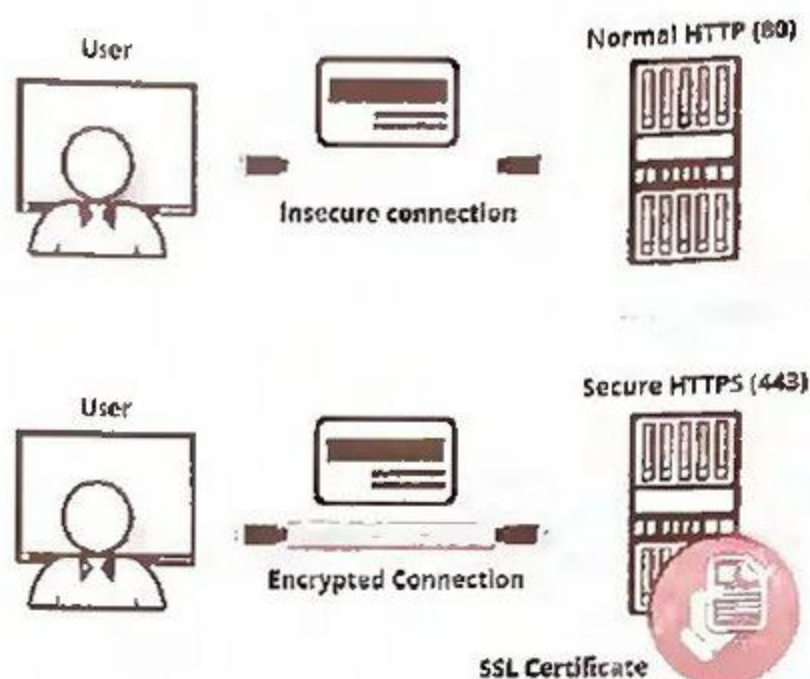


Fig. 12.10: HTTP vs HTTPS

12.11 IDENTITY VERIFICATION

An identity verification service is used to ensure that users or customers provide information that is associated with the identity of a real person. The service may verify the authenticity of physical identity documents such as a driver's licence or passport, called documentary verification, or may verify identity information from authoritative sources such as a credit bureau or government data, called non-documentary verification.

Identity verification service is used both online and in person to verify identity. These services are used by some social networking sites, internet forums and wikis to stop sockpuppetry, underage signups, spamming and illegal activities like harassment, scams and money laundering. These services are required to establish bank accounts and other financial accounts in many jurisdictions.

12.12 CYBER SAFETY IN A NUTSHELL

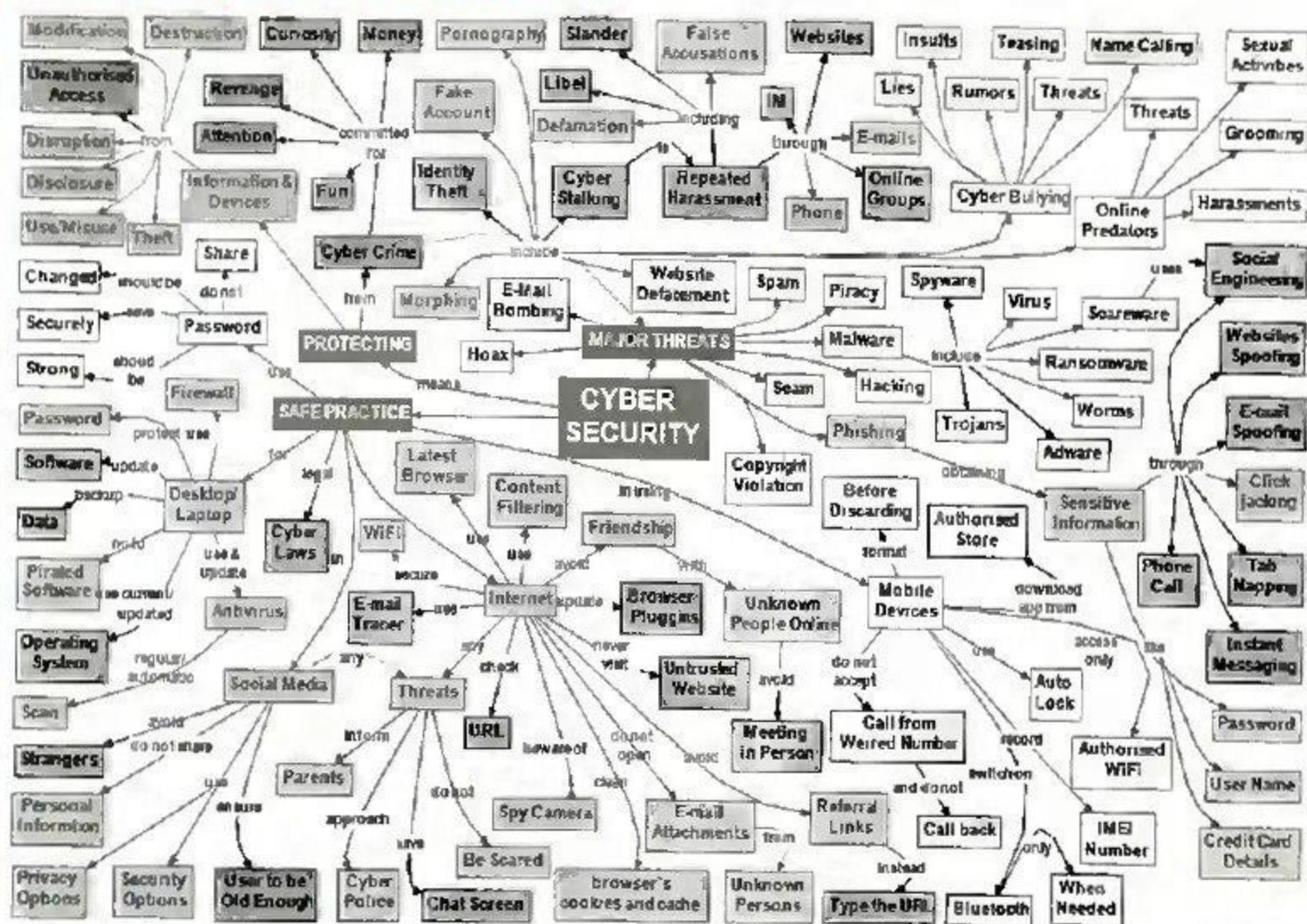


Fig. 12.11: Cyber Safety in a Nutshell

12.13 INFORMATION TECHNOLOGY ACT, 2000

The Information Technology Act, 2000 (also known as ITA-2000, or the IT Act) is an Act of the Indian Parliament (No. 21 of 2000) notified on 17 October, 2000. It is the primary law in India dealing with cybercrime and electronic commerce.

The original Act contained 94 sections, divided into 13 chapters and 4 schedules. The laws apply to the whole of India. Persons of other nationalities can also be indicted under the law if the crime involves a computer or network located in India.

The IT Act is a set of recent legal enactments, currently existing in India, which provide legal support to computer users against cybercrime.

The cyber police works as a detector to discover cybercrimes. They have the authority in respect of all the offences committed under TITA (The Information Technology Act, 2000), Central Act No. 21 of 2000, or crimes related to Intellectual Property Rights. The cyber laws in India and the provisions for legal action and punishment have been explained in a nutshell in Fig. 12.12.



SECTION	OFFENCE	PENALTY
67A	Publishing images containing sexual acts	Imprisonment up to seven years, and/or fine up to ₹ 10,00,000.
67B	Publishing child porn or predating children online	Imprisonment up to five years and/or fine up to ₹ 10,00,000 on 1 st conviction. Imprisonment up to seven years and/or fine up to ₹ 10,00,000 on 2 nd conviction.
67C	Failure to maintain records	Imprisonment up to three years and/or fine up to ₹ 2,00,000.
68	Failure/refusal to comply with orders	Imprisonment up to seven years and/or possible fine.
69	Failure/refusal to decrypt data	Imprisonment up to three years and/or fine up to ₹ 1,00,000.
70	Securing access or attempting to secure access to a protected system	Imprisonment up to ten years and/or fine.
71	Misrepresentation	Imprisonment up to three years and/or fine up to ₹ 1,00,000.

Fig. 12.12: IT ACT, 2000 (Cyber law) of India

IT Act, 2000—Amendments

A major amendment was made to the IT Act in 2008. It introduced Section 66A which penalized sending of "offensive messages". It also introduced Section 69, which gave authorities the power of "interception or monitoring or decryption of any information through any computer resource". Amendments also contained penalties for child pornography, cyberterrorism and voyeurism. The Act was passed in December 2008 and came into force in October 2009.

12.14 UNIQUE IDs AND BIOMETRICS

Biometrics is the science and technology of measuring and analyzing biological data. Biometrics refers to technologies that measure and analyze human body characteristics such as DNA, fingerprints, eye retinas and irises, voice patterns, facial patterns and hand measurements for authentication purposes. It provides the most accurate and unique method of identification.

			
FINGERPRINTS	FACIAL RECOGNITION	IRIS	DNA
• 5–9 seconds processing time • Commonly used in border management • Also used in law enforcement	• Non-invasive collection • Currently used for passports and National ID documents	• Low false acceptance rates • Difficult to replicate • Two-second processing time	• Establishes familial relationship • Commonly used in law enforcement • Highly unique/impossible to replicate

Fig. 12.13: Classification of Biometric Authentication System

Biometric system works by recording and comparing biometric characteristics. When an individual first uses a biometric system, their identifying features are enrolled as a reference for future comparison. This reference may be stored in a central database or on a card (or both) depending on the needs of the application. When biometric recognition is required, the individual's biometric characteristics are recorded again. This time, however, the identifying features are compared by the system with the stored reference to determine if there is a close match.

Aadhaar Card is the best available combination of unique IDs and biometric system as it provides unique identification and authentication through identification number along with thumb impressions and iris patterns.



12.15 IMPACT OF TECHNOLOGY CHANGE ON SOCIETY

Science and technology are essential ingredients of modern life. They transcend local boundaries and touch everyone's life. The technological factors represent conditions created by man that have a profound influence on his life. Technology is a product of civilization. Technology changes society by changing the environment to which we, in turn, adapt. This change is usually in the material environment and the adjustment that we make with these changes often modifies customs and social institutions.

Technology and Industrialization: Technology has contributed to the growth of industries or to the process of industrialization. Industrialization is associated with the factory system of production. Factories have brought down prices of commodities, improved their quality and maximized output. The whole process of production is mechanized. Consequently, traditional skills have declined and a good number of artisans have lost their jobs. Big factories could provide employment opportunities to thousands of people. The process of industrialization has affected the nature, character and growth of the economy. It has contributed to the growth of cities or to the process of urbanization.



Fig. 12.14: Impact of Technology Change

Technology and Urbanization: In many countries, the growth of industries has contributed to the growth of cities.

Urbanization can be described as a process of becoming urban, moving to cities, changing from agriculture to other pursuits common to cities, and corresponding chain of behavioural patterns. As a result of industrialization, people started moving towards industrial areas in search of employment. Due to this, industrial areas developed into towns and cities.

Technology and Unemployment: The problem of unemployment is a concomitant feature of rapid technological advancement. Machines not only provide employment opportunities to men but they also take away their jobs through labour-saving devices. This results in technological unemployment.

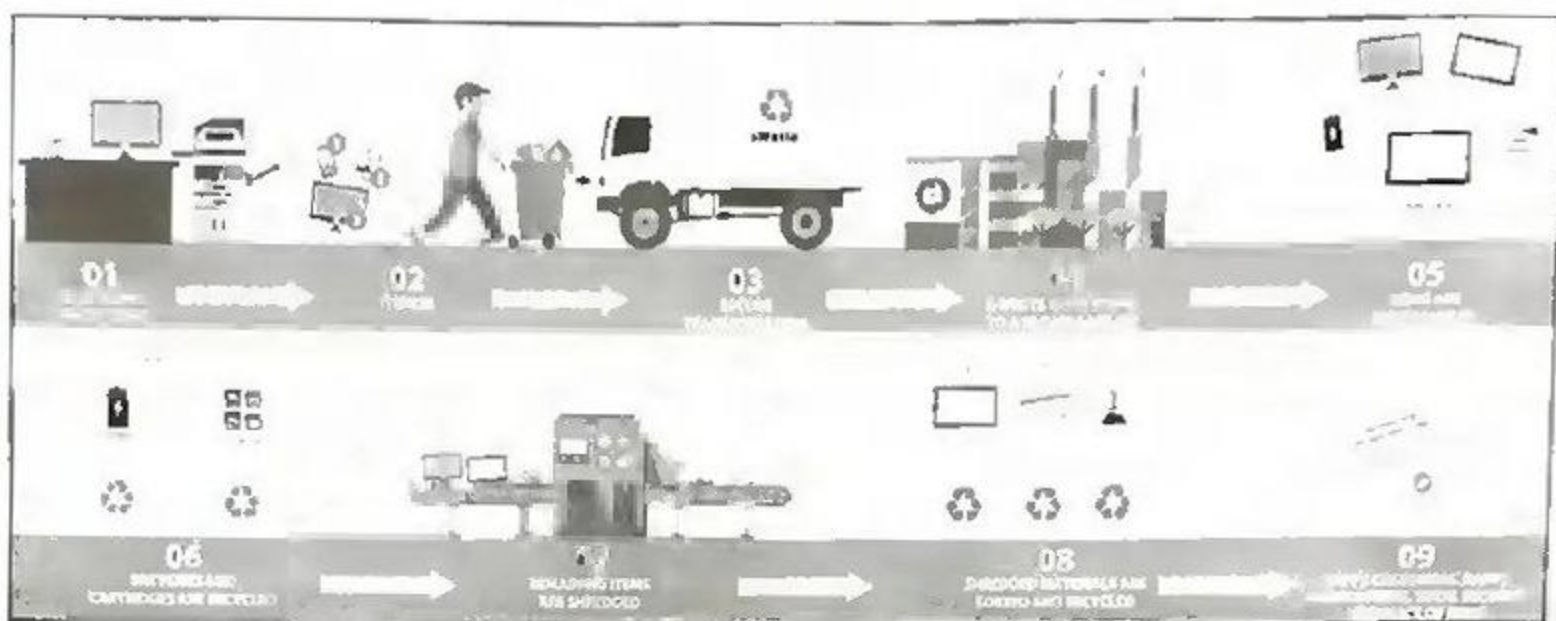
Changes in Social Institutions: Technology has profoundly altered our modes of life. Technology has not spared even the social institutions of its effects. The institutions of family, religion, morality, marriage, state, property have been altered. Modern technology, in taking away industry from the household, has radically changed the family organization. Many functions of the family have been taken away by other agencies. Marriage is losing its sanctity. It is treated as a civil contract than a sacred bond. Marriages are becoming more and more unstable. Instances of divorce, desertion and separation are increasing. Technology has elevated the status of women but it has also contributed to the stresses and strains in relations between man and woman at home. Religion is losing its hold over members. People are becoming more secular, rational and scientific but less religious in their outlook. Inventions and discoveries in science have shaken the foundations of religion. The function of the state or the field of state activity has been widened. Modern technology has made the states perform such functions as the protection of the aged, the weaker sections and the minorities, making provisions for education, health care, etc.

12.16 E-WASTE MANAGEMENT

All electronic waste is made up of deadly chemicals such as lead, cadmium, beryllium, mercury and brominated flame retardants. Disposing of gadgets and devices improperly increases the chances of these dangerous chemicals contaminating the soil, polluting the air and leaching into water bodies.



When e-waste is deposited in a landfill, it tends to leach when water passes through it, picking up trace elements. The contaminated landfill water then reaches natural groundwater with increased toxic levels and can be very harmful if it enters any drinking water source.



Despite having an eco-friendly approach, recycling usually leads to overseas shipping and dumping the gadgets in pits. Worse still, some recycling companies ship e-waste to third world countries.

A lot of children in such countries earn their livelihoods by scavenging gold, silver, iron and copper from the tech waste which is harmful to their health. Countries that are used as dumping grounds usually have high rates of cybercrime as the salvaged hard drives can give criminals direct access to your personal files and information.



Here are some eco-friendly waste disposal techniques that you can use to dispose of electronic waste locally:

➤ Give Your Electronic Waste to a Certified E-Waste Recycler

The positive aspect of e-waste recycling is that you have quite a few recycling options. You need to find an e-waste recycler which is officially certified by the Basel Action Network (BAN). BAN is a non-profit organization of recycling companies dedicated to recycling e-waste in a safe and responsible way. All members have to make a pledge and display their Pledge of Responsible Recycling. So, working alongside a certified recycler means that you don't have to worry about polluting another nation or risk losing your personal details to criminals.

Precautions to be taken before donating or recycling your electronics:

- Upgrade your computer instead of simply replacing it.
- Format all your personal information from your product before discarding it.
- Take out the batteries from your gadgets before getting rid of them.

➤ **Sell off Your Outdated Technology**

As the old saying goes, 'One man's junk is another man's treasure'. This can be applied to helping you get rid of your old electronics. You can tap into online sites like craigslist, eBay or even resort to having a garage sale as this will help you get rid of your outdated electronics as well as earn some money.

➤ **Donating Your Outdated Technology**

Old gadgets that you no longer need can be donated as they may be useful to others. Your old computer may be useful to either an NGO or a student. You should ask yourself these two questions before disposing of your old electronics:

- Is the electronic item working?
- Does the computer have any of your personal information?

A lot of organizations and businesses offer electronic donation programs which you can choose from.

➤ **Visit Civic Institutions**

Enquire amongst your government, universities and schools for any recycling programs they run as a lot of organizations have started assigning a certain day and place for environmentally conscious citizens to come and drop off their e-waste.

➤ **Give Back to Your Electronic Companies and Drop-off Points**

A lot of electronic companies tend to have an exchange policy whereby they take back your old gadgets when you buy a later version, sometimes offering you a discount on your new purchase. A few recycling companies have set up electronic drop-off initiatives along with drop-off points for products such as cell phones and tablets after which they are recycled. You can ask your local electronics shops regarding any information about drop-off locations.

➤ **Safeguard both the Environment and Your Sensitive Information**

Electronics are an important part of our lives today but the flip side is the e-waste that comes with it. So, make sure to format your electronic devices before disposing them of in a proper manner as the consequences of not doing so can be disastrous.

12.17 GENDER AND DISABILITY ISSUES WHILE TEACHING AND USING COMPUTERS

While there is recognition of the potential of Information and Communication Technologies (ICT) as a tool for the promotion of gender equality and empowerment of women, a "gender divide" has also been identified, reflected in the lower number of women accessing and using ICT as compared to men. Unless this gender divide is specifically addressed, there is a risk that ICT may exacerbate existing inequalities between women and men and create new forms of inequality. If, however, the gender dimensions of ICT—in terms of access and use, capacity-building opportunities, employment and potential for empowerment—are explicitly identified and addressed, ICT can be a powerful catalyst for political and social empowerment of women, and for promotion of gender equality.

Disability Issues

Clinical categories of cognitive disabilities include autism and Down's syndrome. Less severe cognitive conditions include the sub-category of so-called learning disabilities, such as dyslexia (reading) and dyscalculia (mathematics). The functional disability perspective ignores the medical and behavioural causes of cognitive disabilities and focuses instead on the abilities and challenges a person with a cognitive disability faces.

Functional cognitive disabilities may involve difficulties or deficits involving:

- Problem-solving
- Attention
- Memory
- Math comprehension
- Visual comprehension
- Reading
- Linguistic (speech)
- Verbal comprehension

The following list shows the benefits that access to ICTs can bring to people within the wide spectrum of learning disabilities. These include:

- **Improved writing**—Standard word processors contain built-in tools for checking grammar, spell-check and predictive typing. Specialized writing support programs, such as Clicker 5, can aid word recognition and writing through the use of symbols or pictures and speech supported grids.
- **Multimedia**—The use of multimedia such as graphics, sound and video can stimulate and encourage interaction and some degree of learning for people with more profound cognitive impairments as well as for pre-literate children. Much of the software required to create multimedia is freely available online.
- **Sensory stimulation**—The use of switches, combined with specialized software games, can enable some people with profound and multiple learning difficulties to play basic cause-and-effect games and even develop some basic computer interaction skills. In addition to these computer-based activities, the use of augmentative and alternative communication strategies and devices, particularly those employing symbols, can aid communication for persons with more profound and multiple-learning impairments.



MEMORY BYTES

- Cyber Safety refers to the safe and responsible use of internet to ensure safety and security of personal information and not posing threat to anyone else's information.
- Use strong unique passwords online.
- Cookies are small text files—bits of information—left on your computer by websites you have visited which let them 'remember' things about you.
- While using social media, one needs to take care of security through password, be aware of fake information and be careful while befriending unknown persons.
- Care must be taken while sharing anything on social media as it may create havoc if mishandled, particularly our personal sensitive information.

- Identity theft occurs when someone, without your knowledge, acquires a piece of your personal information and uses it to commit fraud.
- Confidentiality of information signifies that only authorized users get access to sensitive and crucial information.
- Cyber trolling is an act of cybercrime where a person intentionally starts arguments or upsets others by posting provocative comments.
- A Trojan horse is a program which is intended to carry out malicious operations without the knowledge of the user.
- A worm is a computer program which copies itself across a network.
- Phishing is the attempt to acquire sensitive information such as usernames, passwords, credit card details, etc.
- While entering any sensitive personal information on a website, look for HTTPS in the address bar to make sure your data is encrypted.
- An identity verification service is used to ensure that users or customers provide information that is associated with the identity of a real person.
- You should set up privacy settings for a social media site on your own.
- Digital footprint is the trail of data we leave behind when we visit any website (or use any online application or portal) to fill-in data or perform any transaction.
- All electronic waste is made up of deadly chemicals such as lead, cadmium, beryllium, mercury and brominated flame retardants.

OBJECTIVE TYPE QUESTIONS

1. Fill in the blanks.

- (a) is a potential violation of security.
- (b) Stealing of information is an example of
- (c) A self-replicating program that eats up the entire disk space or memory is called
- (d) is a software which is installed on your computer to spy on your activities.
- (e) A is a program that appears harmless but actually performs malicious functions.
- (f) refers to sending of bulk mails by an identified or unidentified source.
- (g) Most hackers crack of system accounts and gain entry into remote computer systems.
- (h) When we visit certain websites with advertisements, the ads create that store and track our online patterns.
- (i) Ensuring that users or customers provide information that is associated with identity of a real person is known as verification.
- (j) If the user is able to provide a legal email id, he is considered a/an user.
- (k) Unauthorized monitoring of other people's communication is known as
- (l) Flooding a system with junk mail is a type of attack.
- (m) A is a type of browser cookie that is designed to be permanently stored on a user's computer and difficult to detect and remove.
- (n) Discarded electronic devices are called

2. State whether the following statements are True or False.

- (a) Computer security is protecting the integrity, availability and confidentiality of information system.
- (b) Identity theft is a type of fraud.
- (c) Browsing through VPN is anonymous browsing.
- (d) Persistent cookies are different from super cookies.
- (e) Posting something leaves behind traces which are called digital footprints.
- (f) A person who posts derogatory messages online targeting people is called a cyber troll.

- (g) Online theft refers to an act of stealing someone's personal information.
- (h) A firewall only monitors incoming communication.
- (i) To avoid tracking by website, we should browse internet privately.
- (j) We can share sensitive information on wireless networks.
- (k) In cyberbullying, a victim is subjected to a barrage of online messages and emails.
- (l) Spreading rumours online is a punishable offence.
- (m) E-waste is very hazardous if not handled carefully.

3. Multiple Choice Questions (MCQs):

- (a) This term refers to safe and responsible use of technology.
 - (i) Cyber user (ii) Cyber present (iii) Cyber safety (iv) Cyber logic
- (b) This is a type of fraud that involves using someone else's identity to gain other benefits.
 - (i) Personal theft (ii) Information theft (iii) Cyber theft (iv) Identity theft
- (c) These are the small text files for storing a small piece of information related to someone's online habits.
 - (i) Spyware (ii) Virus (iii) Cookies (iv) Spam
- (d) This is a unique address assigned to every computer when connected to a network.
 - (i) MAC address (ii) Computer name (iii) URL address (iv) IP address
- (e) This term ensures that only authorized users get access to sensitive and protected data.
 - (i) Safety (ii) Protection (iii) Confidentiality (iv) Provide
- (f) This is a criminal offence that is facilitated by, or involves the use of, electronic communications or information system, including any electronic device.
 - (i) Network crime (ii) Internet crime (iii) Cybercrime (iv) Super crime
- (g) Which of the following is not an external threat to a computer or a computer network.
 - (i) Ignorance (ii) Trojan horse (iii) Adware (iv) Crackers
- (h) When a person is harassed repeatedly by being followed, called or written to, he/she is a target of:
 - (i) Bullying (ii) Identity theft (iii) Stalking (iv) Phishing
- (i) What is the name of the IT law that India is having in the Indian legislature?
 - (i) Indian Technology (IT) Act, 2000 (ii) Digital Information Technology Act, 2000
 - (iii) Information Technology (IT) Act, 2000 (iv) The Technology Act, 2000
- (j) This term refers to a person who purposely posts opposing, sarcastic, demeaning or insulting comments about something or someone.
 - (i) Cyber criminal (ii) Cyber thief (iii) Cyber troller (iv) Cyber stalker
- (k) These are the records and traces an individual leaves behind as they use the internet.
 - (i) Digital path (ii) Digital map (iii) Digital foot (iv) Digital footprints
- (l) Unauthorized monitoring of other people's communication is called:
 - (i) Spamming (ii) Malware (iii) Adware (iv) Eavesdropping

SOLVED QUESTIONS

1. What is an IP address?

Ans. IP address allows a computer or any other digital device to communicate with someone through the internet. An internet protocol is a set of rules that governs Internet activity and facilitates a variety of actions on the World Wide Web.

An IP address consists of four numbers, each of which contains one to three digits, separated by a single dot (.). Each of the four numbers can range from 0 to 255.

Example of an IP address: 78.125.0.209. Every machine that is on the Internet has a unique IP number.

2. What is social networking?

Ans. Social networking is the practice of expanding the number of one's business and/or social contacts by making connections through individuals.

3. Write down names of any three social networking sites.

Ans. Facebook, Twitter, LinkedIn, Google+ (any three).

4. What are the merits of social networking?

Ans.

- Lowest cost form of marketing
- Huge potential audience and the possibility of messages going viral
- Offers a closer connection with clients
- Source of instant feedback

5. What are the demerits of social networking?

Ans.

- Unreliable information
- Lack of control
- Can be addictive

6. What is Cyber Trolling?

Ans. The internet troll is a modern version of the mythological version. They hide behind their computer screens and actively go out of their way to cause trouble on the internet.

7. Write down the medium/ways of trolling.

Ans. YouTube video comments, Blog comments, Forums, Email, FB, Twitter, Instagram, Social Networking sites and Anonymous ways of networking.

8. What is Cyber stalking?

Ans. Cyber stalking is defined as the unlawful act of harassing a person or collecting an individual's private information using electronic network.

9. Name three data security concepts.

Ans. The three concepts are: encryption, user authentication and data backup.

10. What is personal information?

Ans. Personal information is the information that identifies you or could identify you, *for example*, your name or address.

11. What is meant by the term cyber forensics?

Ans. Cyber forensics is an electronic discovery technique used to determine and reveal technical criminal evidence. It often involves electronic data storage extraction for legal purposes. Although still in its infancy, cyber forensics is gaining traction as a viable way of interpreting evidence. Cyber forensics is also known as computer forensics.

12. What is Information Technology security?

Ans. IT Security is a term which is concerned with the protection of hardware, software and a network of an organization, from the perils of disaster and external attacks (through virus, hacking, etc.).

13. What do you mean by a hacker?

Ans. A hacker is an expert computer programmer who enjoys finding out the inner workings of computer systems or networks. Some have a reputation of using their expertise to illegally break into secure programs in computers hooked up to the internet or other networks.

14. Which of the following crime(s) is covered under cybercrime?

- (a) Stealing a brand new hard disk from a shop.
- (b) Getting into an unknown person's social networking account and messaging on his behalf.
- (c) Copying some important data from a computer without taking permission from the owner of the data.

Ans. (b) and (c)

15. What is Social Media?

Ans. The term social media refers to web and mobile technologies and the practice to share content, opinions, insights, experiences and perspective online. There are many prominent examples of social media platforms, including Facebook, Twitter, YouTube, LinkedIn and blogging sites, among others.

16. Mr Jayanto Das is confused between Twitter and LinkedIn. With the help of examples, explain the difference between them.

Ans. (a) Twitter allows you to have "followers". In short, you are each other's spectators and content creators. A user can post a "tweet" which is usually a short text post, and others can comment, like or "retweet" it and share it. It is common to add a hashtag (#yourhashtaghere) when you want a bigger audience to see your tweet or you are talking about a common topic (something like #GameofThrones, #Netflix, etc). Twitter proved especially important for the Arab Spring and other movements. So, it is useful to pay attention to trending hashtags to be in the know. However, your posts have character limits, meaning you can't write as many words as you like. The current character limit is 280. Users may write using abbreviations such as "sry" (sorry) to capitalize on their tweets.

(b) LinkedIn is generally used as a means of connecting people looking to conduct business. A typical profile can include your CV, credentials, your recent publications and your connection of colleagues. It is one way that people seek employment or collaboration, simply based on connections or common interests. You can choose to pay for a Premium account – they will inform you if managers or recruiters have looked at your profile. They will also give you insight into skills being sought after by recruiters so that you can acquire them as soon as possible and stand out.

17. Define five methods to prevent identity theft.

Ans. To prevent identity theft—

- Never share your password or account numbers over email or instant messaging.
- Do not follow links from emails while conducting financial transactions; instead enter the URL yourself.
- Be wary of callers, pop-ups, websites, or emails asking for personal information.
- Look for "https" or a picture of a lock before purchasing from a website or giving out information on that website. Create a secure password and change it often.
- Install firewall, anti-spyware and anti-virus software and update it often.

18. Why do people troll?

Ans. People troll to annoy others but when brands are involved, the situation reaches another level. Trolling gives the trolls a temporary feeling of power, a much-needed ego boost and makes them look "cool".

19. What is the purpose of web browser?

Ans. Web browser is used to run the software application that allows retrieving, presenting and traversing information from one place to another.

- Web browser provides resources using the WWW (World Wide Web). This can be identified by URI (Uniform Resource Identifier).
- It fetches data like web page, image, video or other piece of content from the server and displays it accordingly.
- It uses hyperlinks to display resources and allows users to navigate their browsers according to resources.
- Web browser defines the application software that is designed for the user to access and retrieve the documents using the internet.

20. What are the privacy and security features being provided by web browsers?

Ans. Web browser supports HTTP secure method that allows more security for the sites that are involved on a day-to-day basis transaction. Security features include—

- Quick and easy way to remove unwanted material or things that are stored without permission like cache, cookies and history.
- Use of blockers to prevent unwanted access to websites from outside.
- A web browser uses other features like disabling of JavaScript and pop-up blockers to block any unwanted scripts or messages to run on the system.
- It also addresses security concerns with regard to blocking access to secure sites that provide money transfer facilities, etc.

21. What is the support being provided by web browser?

Ans. A web browser supports the HTML version that is used to create the website in a very simple manner without using complex tools.

- Web browser supports rapid development of the websites and tools that can be used for the creation of it.
- It provides a way to develop non-standard dialects of HTML that provide interoperability support.
- Web browser provides standard libraries through which the support can be given to make standard-based HTML pages.
- It provides support for other languages like JavaScript, HTML or XHTML that can be rendered by the web browsers.

22. What is cyber stalking?

Ans. Cyber stalking is defined as the unlawful act to harass a person, or collecting an individual's private information using electronic network.

23. Explain gender and disability issues while teaching and using computers.

Ans. Studies have shown that in many countries computer use in schools is dominated by men. Female teachers have less regard for their own skills and knowledge than their male counterparts. Females know less about information technology, enjoy using the computer less than male students, and perceive more problems with software.

24. What are the possible causes that contribute to gender and disability issues? [HOTS]

Ans. Possible causes that contribute to this problem are:

- Differences in parental support.
- Restricted access to computers.
- Fewer female role models and a casual approach towards computer activities in school.

Gender differences are being found both inside and outside the school. This means that both teachers and parents have to be made aware of this before remedial measures are initiated. Same is the case with people with disabilities.

25. Describe some common forms of student plagiarism.

Ans. According to 'The Reality and Solution of College Plagiarism' created by the Health Informatics Department of the University of Illinois, Chicago, mainly there are 10 forms of plagiarism that the students commit:

- Submitting someone else's work as their own.
- Taking passages from their own previous work without adding citations.
- Rewriting someone else's work without properly citing sources.
- Using quotations without citing the source.
- Interweaving various sources together in the work without citations.
- Citing some passages, and not all, that should be cited.
- Melding together cited and uncited sections of the work.
- Providing proper citations without much changing the structure and language of the borrowed ideas.
- Citing the source inaccurately.
- Relying too much on others' works and failing to bring original ideas into the text.

26. What do you understand by Biometric Authentication?

Ans. Biometrics is the technical term for body measurements and calculations. It refers to metrics related to human characteristics. Biometric authentication (or realistic authentication) is used in Computer Science as a form of identification and access control. It is also used to identify individuals in groups that are under surveillance.

27. Ms Sharmishtha has many electronic gadgets which are not usable due to outdated hardware and software. Help her to find three best ways to dispose of used electronic gadgets.

- Ans.
1. Give your electronic waste to a certified e-waste recycler
 2. Donating your outdated technology.
 3. Give back to your electronic companies or at drop-off points.

28. Describe measures to recycle your e-waste safely.

Ans. The following measures can be adopted to recycle the e-waste safely:

1. **Use a certified e-waste recycler:** Find an e-waste recycler certified by the Basel Action Network (BAN). BAN is a non-profit organization devoted to certifying e-Stewards, recyclers who are committed to safely and responsibly recycling electronics. Members take and demonstrate the Pledge of Responsible Recycling, so working with a certified e-Steward means you don't have to worry that your gadget will become another nation's pollution or a criminal's newest project. BAN's recycler locator will help you find the certified safety and comfort of e-Stewards in your area.
2. **Visit Civic Institutions:** Check with your local government, schools, and universities for additional responsible recycling options. With e-waste becoming such a large problem, government offices and schools are assigning days when citizens can bring unwanted electronics to a designated drop-off location.
Many communities post a calendar that will include recycling days, so check your local paper or visit their website. When you recycle your items locally, you can make the occasion a day trip and a community event. Encourage your neighbours to join you and spread the word about educated e-waste disposal.
3. **Explore Retail Options:** Best Buy, for example, isn't certified through BAN's e-Steward program, but they do have an effective recycling program in all of their stores. They claim to use only recyclers which adhere to the highest standards of e-waste processing. Specifically, their website discloses that the e-waste that you bring to their stores will not end up in a foreign country or in any landfill: "We partner directly with a short list of qualified, respected recycling companies who ensure all products collected for recycling through Best Buy are handled responsibly. These recycling companies meet our standards, and we encourage them to examine and consider additional third party standards for responsible practices (such as the EPA R2 and e-Stewards)."

You can drop off all kinds of e-waste for recycling at Best Buy. These include:

- Cell phones
- Power cords
- Speakers
- Paper shredders
- Desktops
- Notebooks
- TVs
- GPS devices
- DVD players
- Memory cards
- Laptops

4. **Donate your electronics:** Reusing is always better than recycling. If your electronics still have life left, you can reduce e-waste pollution and share technology with people who wouldn't otherwise have access to it.

29. List the key features that make Biometric systems most authenticated and trusted.

Ans. A number of factors are weighed before selecting a particular biometric for use in a specific application.

They include:

- **Universality:** It means that every person using a system should possess the trait.
- **Uniqueness:** This means the trait should be sufficiently different for individuals in the relevant population so that they can be distinguished from one another.
- **Permanence:** It relates to the manner in which a trait varies over time. More specifically, a trait with 'good' permanence will be reasonably invariant over time with respect to the specific matching algorithm.
- **Measurability (Collectability):** This relates to the ease of acquisition or measurement of the trait. In addition, acquired data should be in a form that permits subsequent processing and extraction of the relevant feature sets.

- **Performance:** It relates to accuracy, speed and robustness of technology.
- **Acceptability:** This relates to how well individuals in the relevant population accept the technology so that they are willing to have their biometric trait captured and assessed.
- **Circumvention:** It relates to the ease with which a trait might be imitated using an artefact or substitute.

30. How can you use computers and internet to plan a vacation? Name a popular website that you think can be helpful.

Ans. We can book tickets, hotels, cabs and even visits to historical places where tickets are required. We can plan our itinerary and make appropriate reservations through makemytrip.com, etc.

UNSOLVED QUESTIONS

- Should we use incognito window during surfing?
- State whether True or False.
 - IP address helps trace our geographical area.
 - Cookies store our online habits.
 - Using strong internet passwords does not help in maintaining cyber safety.
- How can one safely browse the internet?
- What are safe browsing tips?
- Is Google safe for browsing?
- Do you feel safe on the internet?
- Define cyber safety.
- Write at least four tips of cyber safety.
- What is private browsing?
- What are the possible dangers if cyber safety is overlooked?
- What is the role of IP Address in networking?
- What do you mean by confidentiality of information?
- What is the role of firewall?
- What do you mean by cybercrime?
- Name some common social networking sites.
- Define digital footprint.
- Define cookies.
- How do websites track us?
- Robin searched the net to purchase a wrist watch and a football. Now, whenever he goes online, he is flooded with advertisements about sports items and wrist watches.
 - Why does this happen?
 - How could Robin avoid this?
 - How can Robin get rid of this?
- What do you mean by virus?
- What do you mean by spyware and adware?
- What is the difference between spyware and adware?
- What do you mean by malware?
- What do you mean by worm?
- What do you mean by Trojan horse?
- What is the difference between worm and Trojan horse?
- Explain the difference between malware and worm.

28. Why do we need a secure connection while doing netbanking?
29. Is WhatsApp secured?
30. How will you identify from a URL whether the website you are visiting is safe or not?
31. Define secure connection.
32. Differentiate between http and https.
33. Name any two websites which use https.
34. What should we practise to ensure confidentiality of information?
35. Describe the following cybercrimes: Cyberbullying and Cyber Stalking.
36. What do you mean by identity theft? How can we prevent it?
37. What is cyber law?
38. How is cyber law implemented in India?
39. What is IT Act, 2000?
40. What is child pornography?
41. How does cybercrime affect our daily life?
42. Analyze two cybercrime prevention and protection methods.
43. What steps should be taken to protect oneself from online fraud?
44. Write four steps to combat cybercrime.
45. How is social networking responsible for cybercrime?